

Logistics Engineering – Exercise Sheet: Transfer Functions, Feedback, and Replenishment Dynamics

Bachelor in Logistics Engineering

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Exercises on Transfer Functions, Step Response, Feedback, and Replenishment Models

Use zero initial conditions throughout unless stated otherwise. In transfer-function and feedback exercises, $r(t)$ denotes the reference input. In replenishment-balance exercises, $r(t)$ denotes the release rate whenever explicitly stated. Use

$$G(s) = \frac{Y(s)}{R(s)}, \quad T(s) = \frac{G(s)C(s)}{1 + G(s)C(s)H(s)}, \quad u(t) = \text{unit step function.}$$

For stable systems, the static gain is $G(0)$. When useful, use

$$\mathcal{L}^{-1} \left\{ \frac{1}{s+a} \right\} = e^{-at} u(t), \quad \mathcal{L}^{-1} \left\{ \frac{1}{s(s+a)} \right\} = \frac{1}{a} (1 - e^{-at}) u(t),$$

and the standard second-order form

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}.$$

In the replenishment exercises, $I(t)$ denotes inventory deviation, $B(t)$ backlog, $W(t)$ work in process, $d(t)$ demand rate deviation, $q(t)$ replenishment completion rate deviation, $c(t)$ completion rate, and $I^*(t), W^*(t)$ target values. Use Little's law

$$WIP \approx TH \cdot CT$$

whenever requested. When numerical approximations are needed, round to three decimals. Leave exponential expressions in exact form whenever convenient.

1. **Ex 1 :** A warehouse throughput dynamics is modeled by

$$3\dot{y}(t) + y(t) = 6r(t).$$

- (a) Take the Laplace transform and determine the transfer function

$$G(s) = \frac{Y(s)}{R(s)}.$$

(8 Points)

- (b) Identify the static gain K and the time constant τ . (6 Points)

- (c) If the throughput command is

$$r(t) = 2u(t),$$

determine the long-run value y_∞ . (6 Points)

2. **Ex 2 :** An automated shuttle position loop is described by

$$\ddot{x}(t) + 4\dot{x}(t) + 13x(t) = 13r(t).$$

- (a) Determine the transfer function

$$G(s) = \frac{X(s)}{R(s)}.$$

(8 Points)

- (b) Compare the result with the standard second-order form and determine

$$\omega_n \quad \text{and} \quad \zeta.$$

(8 Points)

- (c) Classify the system as overdamped, critically damped, or underdamped. (4 Points)

3. **Ex 3 :** Two transfer functions are used for logistics equipment models.

- (a) For

$$G_1(s) = \frac{2}{s + 5},$$

determine the order, the poles, the zeros, and the static gain $G_1(0)$.

(8 Points)

(b) For

$$G_2(s) = \frac{s+2}{s^2+4s+8},$$

determine the order, the zero, the poles, and the static gain $G_2(0)$.
(12 Points)

4. **Ex 4 :** Consider the following three transfer functions:

$$G_A(s) = \frac{1}{(s+1)(s+3)}, \quad G_B(s) = \frac{1}{s-2}, \quad G_C(s) = \frac{5}{s^2+6s+13}.$$

- (a) Determine the poles of all three systems. (8 Points)
- (b) Classify each system as stable or unstable. (8 Points)
- (c) Which system has oscillatory decay, and why? (4 Points)

5. **Ex 5 :** A packaging station has transfer function

$$G(s) = \frac{4}{2s+1}$$

and receives the step input

$$r(t) = 3u(t).$$

(a) Compute

$$Y(s) = G(s)R(s).$$

(6 Points)

(b) Decompose $Y(s)$ into partial fractions of the form

$$\frac{A}{s} + \frac{B}{2s+1}.$$

(8 Points)

(c) Determine the time response $y(t)$ and the final value. (6 Points)

6. **Ex 6 :** A conveyor-speed adjustment is modeled by

$$G(s) = \frac{5}{4s+1}.$$

The system receives a unit step input.

- (a) Determine the step response $y(t)$. (6 Points)
- (b) Determine the final value y_∞ and the value $y(4)$. (8 Points)

- (c) Determine the time t_{90} for which

$$y(t_{90}) = 0.9 y_{\infty}.$$

Leave the answer as a logarithm or give a decimal approximation.
(6 Points)

7. **Ex 7 :** A positioning system has transfer function

$$G(s) = \frac{36}{s^2 + 12s + 36}.$$

- (a) Determine

$$\omega_n \quad \text{and} \quad \zeta.$$

(8 Points)

- (b) Compute the poles. (8 Points)

- (c) State the qualitative type of the step response. (4 Points)

8. **Ex 8 :** A shuttle-position model is given by

$$G(s) = \frac{4}{s^2 + 2s + 4}.$$

For a unit step input:

- (a) Compute

$$Y(s) = \frac{4}{s(s^2 + 2s + 4)}.$$

(6 Points)

- (b) Using partial fractions and completion of the square, show that

$$Y(s) = \frac{1}{s} - \frac{s+1}{(s+1)^2 + 3} - \frac{1}{(s+1)^2 + 3}.$$

(8 Points)

- (c) Determine the time response $y(t)$. (6 Points)

9. **Ex 9 :** A throughput regulation loop uses unity feedback with

$$G(s) = \frac{1}{3s+1}, \quad C(s) = 4.$$

- (a) Determine the closed-loop transfer function

$$\frac{Y(s)}{R(s)}.$$

(8 Points)

- (b) Determine the closed-loop pole and the closed-loop time constant. (6 Points)
- (c) Determine the DC gain and the steady-state error for a unit step reference. (6 Points)

10. **Ex 10 :** A controlled logistics system uses non-unity feedback with

$$G(s) = \frac{2}{s+2}, \quad C(s) = 3, \quad H(s) = 2.$$

- (a) Determine the closed-loop transfer function

$$\frac{Y(s)}{R(s)}.$$

(10 Points)

- (b) Determine the closed-loop pole and the DC gain. (6 Points)
- (c) State whether perfect step tracking occurs. (4 Points)

11. **Ex 11 :** A unity-feedback loop is used to reject output disturbances. The plant and controller are

$$G(s) = \frac{1}{2s+1}, \quad C(s) = 5.$$

An additive disturbance $D(s)$ enters at the output.

- (a) Starting from

$$Y(s) = G(s)C(s)(R(s) - Y(s)) + D(s),$$

derive the transfer function

$$\frac{Y(s)}{D(s)}$$

for the case $R(s) = 0$. (8 Points)

(b) For

$$D(s) = \frac{3}{s}, \quad R(s) = 0,$$

compute $Y(s)$. (6 Points)

(c) Determine the steady-state output deviation caused by this disturbance. (6 Points)

12. **Ex 12 :** A warehouse throughput plant is

$$G(s) = \frac{1}{4s + 1}$$

with unity feedback and proportional controller

$$C(s) = K_c.$$

(a) Derive the general closed-loop transfer function

$$T(s) = \frac{Y(s)}{R(s)}$$

in terms of K_c . (8 Points)

(b) For

$$K_c = 1,$$

determine the closed-loop pole and the steady-state error for a unit step reference. (6 Points)

(c) Repeat part (b) for

$$K_c = 4,$$

and compare the two controllers with respect to speed and step-tracking accuracy. (6 Points)

13. **Ex 13 :** Three conservation equations are used in logistics dynamics:

$$\dot{I} = q - d, \quad \dot{B} = d - q, \quad \dot{W} = r - c.$$

(a) Match each equation to the most natural state variable: inventory, backlog, or work in process. (6 Points)

(b) At one instant,

$$q = 40, \quad d = 55.$$

Compute $\dot{I}$ and $\dot{B}$, and interpret the signs. (8 Points)

(c) At another instant,

$$r = 30, \quad c = 24.$$

Compute $\dot{W}$ and state what this means for line congestion. (6 Points)

14. **Ex 14 :** A spare-parts store uses a reorder-point logic together with a target-inventory correction law.

(a) Given

$$\bar{d} = 18 \text{ units/day}, \quad L = 4 \text{ days}, \quad SS = 12 \text{ units},$$

compute the reorder point

$$R = \bar{d}L + SS.$$

(6 Points)

(b) Let the replenishment rule be

$$q(t) = 0.5(I^*(t) - I(t))$$

and the inventory balance be

$$\dot{I}(t) = q(t) - d(t).$$

Derive the differential equation for $I(t)$. (8 Points)

(c) For the case $d(t) = 0$, determine the transfer function

$$\frac{I(s)}{I^*(s)}.$$

(6 Points)

15. **Ex 15 :** A warehouse inventory deviation obeys

$$\dot{I}(t) + 0.8 I(t) = 0.8 I^*(t), \quad I(0) = 0.$$

(a) Determine the transfer function

$$\frac{I(s)}{I^*(s)}.$$

(6 Points)

- (b) For the target step

$$I^*(t) = 25u(t),$$

compute the response $I(t)$. (8 Points)

- (c) Determine $I(\infty)$ and $I(\tau)$ with

$$\tau = \frac{1}{0.8}.$$

(6 Points)

16. **Ex 16 :** A base-stock inventory model under demand disturbance is

$$\dot{I}(t) + 2I(t) = -d(t), \quad I(0) = 0.$$

- (a) Determine the transfer function

$$\frac{I(s)}{D(s)}.$$

(6 Points)

- (b) For the demand step

$$d(t) = 6u(t),$$

compute the inventory response $I(t)$. (8 Points)

- (c) Determine the long-run inventory deviation and explain its logistics meaning. (6 Points)

17. **Ex 17 :** A KANBAN loop is modeled by

$$\tau_K \dot{q}(t) + q(t) = d(t), \quad \dot{I}(t) = q(t) - d(t), \quad \tau_K = 3.$$

- (a) Determine the transfer function

$$\frac{Q(s)}{D(s)}.$$

(6 Points)

- (b) Determine the transfer function

$$\frac{I(s)}{D(s)}.$$

(8 Points)

(c) For a demand step of size

$$\Delta d = 10,$$

determine the long-run inventory deviation I_∞ . (6 Points)

18. **Ex 18 :** A KANBAN-controlled stage has effective replenishment time constant

$$\tau_K = 2 \text{ hours}$$

and faces the demand step

$$d(t) = 8u(t).$$

(a) Compute the replenishment completion rate $q(t)$. (8 Points)

(b) Compute the inventory deviation $I(t)$. (6 Points)

(c) The stage uses containers of size

$$C = 20,$$

average demand

$$D = 60 \text{ units/hour},$$

replenishment lead time

$$L = 1.5 \text{ hours},$$

and safety factor

$$\alpha = 0.2.$$

Compute the approximate number of cards using

$$n \approx \frac{DL(1 + \alpha)}{C},$$

and round up to the next integer. (6 Points)

19. **Ex 19 :** A CONWIP-controlled production line uses

$$r(t) = c(t) + 0.25(W^*(t) - W(t)), \quad \dot{W}(t) = r(t) - c(t).$$

(a) Derive the closed WIP equation

$$\dot{W}(t) + 0.25 W(t) = 0.25 W^*(t).$$

(6 Points)

- (b) Determine the transfer function

$$\frac{W(s)}{W^*(s)}.$$

(6 Points)

- (c) For the target step

$$W^*(t) = 120u(t), \quad W(0) = 0,$$

compute the WIP response $W(t)$. (4 Points)

- (d) If the steady throughput at the target WIP is

$$TH = 30 \text{ jobs/day},$$

use Little's law to determine the corresponding cycle time CT . (4 Points)

20. **Ex 20 :** Capstone exercise: compare three standard logistics control models.

- (a) A base-stock loop satisfies

$$\dot{I}_B(t) + I_B(t) = I^*(t), \quad I_B(0) = 0.$$

For

$$I^*(t) = 10u(t),$$

compute $I_B(t)$. (6 Points)

- (b) A KANBAN loop satisfies

$$2\dot{q}_K(t) + q_K(t) = d(t), \quad \dot{I}_K(t) = q_K(t) - d(t), \quad I_K(0) = 0.$$

For

$$d(t) = 4u(t),$$

compute $q_K(t)$ and $I_K(t)$. (6 Points)

- (c) A CONWIP loop satisfies

$$\dot{W}_C(t) + 0.5 W_C(t) = 0.5 W^*(t), \quad W_C(0) = 0.$$

For

$$W^*(t) = 100u(t),$$

compute $W_C(t)$. (4 Points)

- (d) Compare the time constants of the base-stock and CONWIP target-tracking loops and state which one reaches 63% of its final value first. (4 Points)